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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 30th July Evening Shift

JEE Main Question Paper with Answer Key

30 Jul 2022	90	16	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	24 questions
Mathematics	22 questions
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Out of syllabus	16 questions
Answer key	at the end

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Chemistry 24 questions

Q1. MCQ (Single Correct Answer)

When the kinetic energy of an electron is increased nine times, the wavelength of the de-Broglie wave associated with it would become :

(A) One-third (B) Three times (C) Two third (D) One sixth

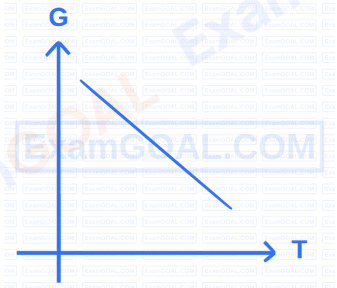
Open solution

Q2. MCQ (Single Correct Answer)

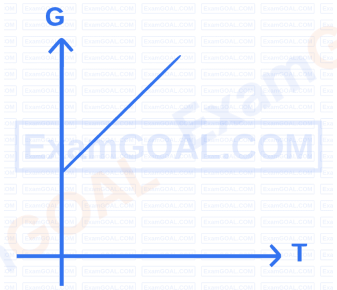
Which of the following is the plot of Gibb's free energy (G) vs temperature (T), at constant pressure, for a pure substance?

Open solution

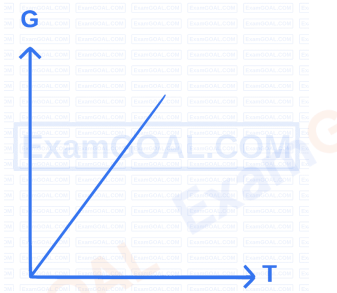
(A)



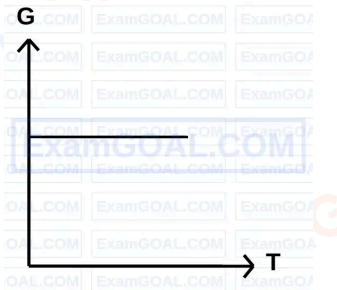
(B)



(C)



(D)



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Q3. MCQ (Single Correct Answer)

For Periodic Table which of the following statements are TRUE

A. Shielding increases as we go down in the group.

B. Shielding increases sharply across the period.

C. Ionization enthalpy increases down the group.

D. Metallic character increases down the group.

E. Electronegativity decreases down the group.

Choose the correct answer from the following options :

(A) B, C and E Only

(B) A, D, and E only

(C) A, C, D and E only

(D) A, D and E only

Open solution

Q4. MCQ (Single Correct Answer)

The correct option for relative covalent nature of compounds is

(A) $KF > KI$

(B) $SnCl_4 > SnCl_2$

(C) $KF > LiF$

(D) $NaCl > HCl$

Open solution

Q5. MCQ (Single Correct Answer)

The E° (volt) values are : Al^{3+}/Al, -1.66, Sc^{3+}/Sc, -2.08, Fe^{3+}/Fe^{2+}, $+0.77$, Hg_2^{2+}/Hg, $+0.79$. Arrangement of cations, Al^{3+}, Sc^{3+}, Fe^{3+} and Hg_2^{2+} in decreasing order of oxidizing strength is

(A) $Al^{3+} > Sc^{3+} > Fe^{3+} > Hg_2^{2+}$

(B) $Hg_2^{2+} > Fe^{3+} > Sc^{3+} > Al^{3+}$

(C) $Hg_2^{2+} > Fe^{3+} > Al^{3+} > Sc^{3+}$

(D) $Sc^{3+} > Al^{3+} > Fe^{3+} > Hg_2^{2+}$

Open solution

Q6. MCQ (Single Correct Answer)

In low and high spin octahedral Co^{3+} complexes the number of electrons in t_{2g} level respectively will be . (given At. no. of Co, 27)

(A) 6 and 3

(B) 6 and 4

(C) 3 and 4

(D) 4 and 6

Open solution

Q7. MCQ (Single Correct Answer)

Glycerol can be separated from spent-lye on industrial scale by using

(A) TLC technique

(B) Distillation under reduced pressure

(C) Differential Extraction

(D) Crystallization

Open solution

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Q8. MCQ (Single Correct Answer)

Open solution

Which of the following is unstable?

(A)



(B)



(C)



(D)

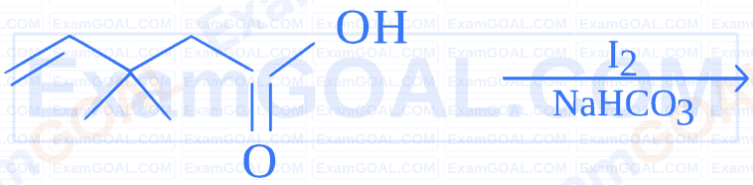


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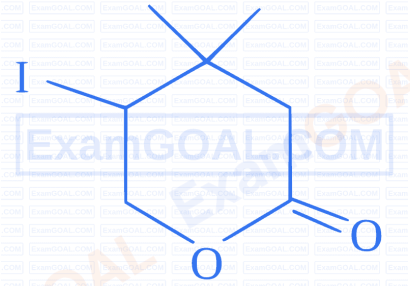
Q9. MCQ (Single Correct Answer)

Open solution

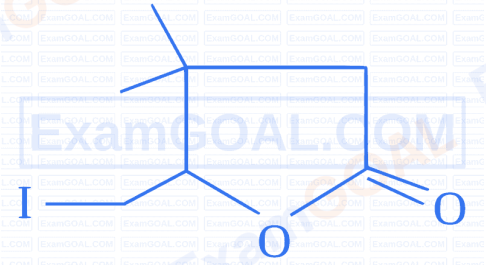
Major product of the following reaction is



(A)



(B)



(C)



(D)



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Q10. MCQ (Single Correct Answer)



Open solution

For the above conversion, the correct sequential addition of reagents is-

(A) (i) Mg (ii) CO_2, dry ether (iii) H_2O

(B) (i) $NaOH$ (ii) C_2H_5OH (iii) HCl

(C) (i) $NaCN$ (ii) H_2O (iii) HCl

(D) (i) $KMnO_4$ (ii) HCl

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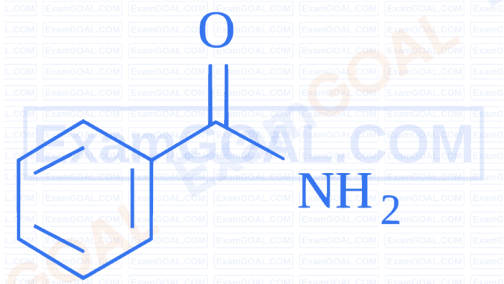
Q11. MCQ (Single Correct Answer)

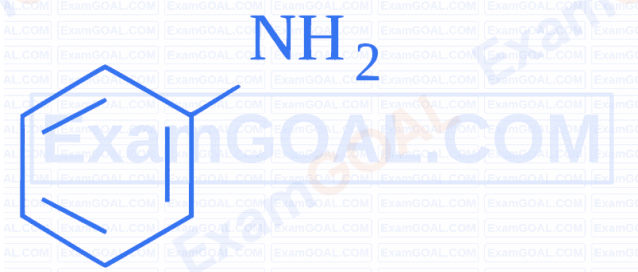
Open solution

Which one of the following compounds can be prepared in good yield by Hoffmann bromamide degradation reaction?

(A) 

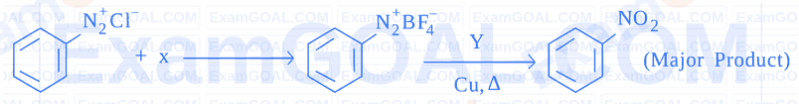
(B) 

(C) 

(D) 

Q12. MCQ (Single Correct Answer)

Open solution



Consider the given chemical reaction, identify X and Y

(A) $X = BF_3 \quad Y = HNO_3$

(B) $X = HBF_4 \quad Y = NaNO_2$

(C) $X = HBF_4 \quad Y = HNO_3$

(D) $X = BF_3 \quad Y = NaNO_2$

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Q13. MCQ (Single Correct Answer)

Open solution

Given below are two statements:

Statement I: Glycogen is highly branched polysaccharide and commonly known as animal starch due to its similarity with amylopectin.

Statement II: Amylose is water insoluble and constitutes about 15-20% of starch.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Q14. MCQ (Single Correct Answer)

Open solution

In the ring test of NO_3^- ion what is the oxidation number of iron in the complex (brown ring) $[Fe(H_2O)_5NO]SO_4$?

(A) $+2$ (B) $+1$ (C) $+3$ (D) $+5$

Q15. Numerical

Open solution

The density of 4 molal solution of NaOH is $1.160\ g\ mL^{-1}$. The molarity of the solution is _____ M. (Given : Molar mass of NaOH $= 40\ g\ mol^{-1}$) (nearest integer)

Q16. Numerical

Open solution

On the basis of MO. theory, the number of molecules from the following which have bond order two, is _____.

O_2, O_2^+, N_2^{2-}, C_2, B_2

Q17. Numerical

Open solution

31 g of ethylene glycol is mixed with 500 g of water. The freezing point of the aqueous solution is _____ K.

(in nearest integer)

(K_f of water $= 1.86\ K\ kg\ mol^{-1}$)

[Molar mass of C, H, O are 12, 1, 16 $g\ mol^{-1}$]

Q18. Numerical

Open solution

50 mL of 0.1 M CH_3COOH is titrated against 0.1 M NaOH solution. When 10 mL of NaOH is added, the pH of the solution becomes _____ $\times 10^{-1}$. (nearest integer)

Given: pK_a (CH_3COOH) = 4.8, log 2 = 0.3

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Q19. Numerical

For the given cell $Zn(s) \mid Zn^{2+}(C_1\ M) \mid\mid Zn^{2+}(C_2\ M) \mid Zn(s)$ the change in Gibbs energy (ΔG) will be zero when $\frac{C_1}{C_2}$ is equal to _____.

Open solution

Q20. Numerical

$X \rightarrow Y + Z$

X decomposes at 700 K to give Y and Z. The results of two measurements are :

Initial concentration of X / (mol/L)	6.0	12.0
Half life / s	1.0	2.0

The rate constant for the reaction is _____ (in appropriate unit)

Open solution

Q21. Numerical

Consider the molecules / ions given below

XeO_3, BF_4^-, I_3^-, SF_6, PCl_5

The ratio of number of molecules having sp^3d hybridisation to those having sp^3 hybridisation is _____

Open solution

Q22. Numerical

The number of coloured and paramagnetic ions from the following in aqueous solution are _____.

Ti^{3+}, Co^{2+}, Ni^{2+}, Cu^{2+}, Cu^{+}, Ti^{4+}, Zn^{2+}, Sc^{3+}

Given Atomic no.

Sc, 21; Ti, 22; Co, 27; Ni, 28; Cu, 29; Zn, 30

Open solution

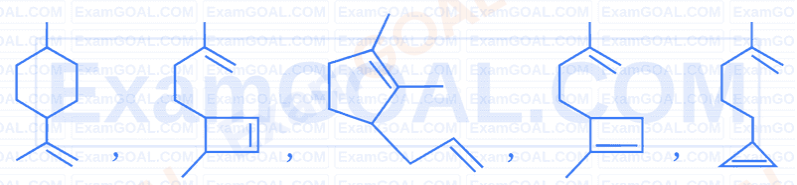
Q23. Numerical

The number of isomeric compounds (structural isomers only) possible with the molecular formula $C_5H_{10}O_2$, which do not react with metallic sodium is _____.

Open solution

Q24. Numerical

Among the following, how many hydrocarbon (s) with formula $C_{10}H_{16}$ which on reaction with acidic potassium permagenate yield(s) same product ? _____



Open solution

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Mathematics 22 questions

Q25. MCQ (Single Correct Answer)

Let $f : \mathbb{R} - \{5\} \to \mathbb{R}$ be defined as $f(x) = \dfrac{2x^2 + 3x - 2}{x - 5}$. Then f is

(A) one-one and onto
(B) one-one but not onto
(C) onto but not one-one
(D) neither one-one nor onto

Open solution

Q26. MCQ (Single Correct Answer)

Let $z_1 = 1 + 2i$, $z_2 = 2 + i$, $\dfrac{1}{z_1} + \dfrac{1}{z_2} = \dfrac{6}{w}$, and $z = \dfrac{iw}{2 - \overline{w}}$.

Consider the statements

(S_1) $|z| = \dfrac{5}{\sqrt{17}}$

(S_2) $\arg(z) + \arg(w) = \tan^{-1}\left(\dfrac{5}{3}\right)$.

Then

(A) both (S_1) and (S_2) are wrong
(B) only (S_1) is correct
(C) only (S_2) is correct
(D) both (S_1) and (S_2) are correct

Open solution

Q27. MCQ (Single Correct Answer)

If the system of linear equations

$\lambda x + y - z = -1$

$x - y - 3z = 2$

$-x + y + z = \mu$

has infinitely many solutions, then the equation of the line passing through the points $(\lambda + 2\mu,\ 2\lambda + \mu)$ and $(1,\ \lambda\mu)$ is

(A) $4x - 6y = -5$
(B) $2x - 2y = 5$
(C) $2x - 2y = -1$
(D) $4x + 6y = 5$

Open solution

Q28. MCQ (Single Correct Answer)

The sum of all values of n, for which ${}^{n-1}C_4 - {}^{n-1}C_3 - \dfrac{5}{4}\,{}^{n-2}P_2 < 0$, is

(A) 40
(B) 45
(C) 55
(D) 56

Open solution

Q29. MCQ (Single Correct Answer)

For $|\alpha| \geq 1$, let $5^{4-2\alpha}$, 63, $5^{2\alpha-1}$ be in A.P. with the common difference d. Let S_n denote the sum of the first n terms of an A.P., whose first term is α^2 and the common difference is d. If $S_{30} - S_{15} = 30k$, then k is equal to

(A) 715
(B) 695
(C) 684
(D) 683

Open solution

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Q30. MCQ (Single Correct Answer)

Open solution

Let

$$f(x)=\begin{cases}\dfrac{\sin(ax+3x)+\sin x}{x}, & x<0\\ 2, & x=0\\ \dfrac{(x+6bx^2)^{\frac{1}{3}}-x^{\frac{1}{3}}}{3x^{4/3}}, & x>0\end{cases}$$

be continuous at $x=0$. If $(1-k)\,f\left(\dfrac{7}{18}\right)=ab$, then k is equal to

(A) -6 (B) 15 (C) -13 (D) 8

Q31. MCQ (Single Correct Answer)

Open solution

If $y(x)=\tan x-\displaystyle\int_0^{2x}(x^2-xt+1)\sin t\,dt$, then at $x=\dfrac{\pi}{3}$, the value of $2y'''-y''$ is equal to

(A) 158 (B) $159-4\sqrt{3}$ (C) 126 (D) $127-4\sqrt{3}$

Q32. MCQ (Single Correct Answer)

Open solution

Let $A_k=\displaystyle\int_{k\pi}^{(k+1)\pi}e^{-x}\sin x\,dx$, $k=0,1,2,\ldots,20$. Then $\displaystyle\sum_{k=0}^{20}|A_k|$ is equal to

(A) $\left(\dfrac{e^{\pi}+1}{e^{\pi}-1}\right)\left(1-e^{-20\pi}\right)$ (B) $\dfrac{1}{2}\left(\dfrac{e^{\pi}+1}{e^{\pi}-1}\right)\left(1-e^{-21\pi}\right)$

(C) $\left(\dfrac{e^{\pi}+1}{e^{\pi}-1}\right)\left(1-e^{-21\pi}\right)$ (D) $\dfrac{1}{2}\left(\dfrac{e^{\pi}+1}{e^{\pi}-1}\right)\left(1-e^{-20\pi}\right)$

Q33. MCQ (Single Correct Answer)

Open solution

If $y=y(x)$ satisfies the differential equation $x\,dy+\left(x\tan\left(\dfrac{y}{x}\right)-y\right)dx=0$, $x>0$, and $y(1)=\dfrac{\pi}{2}$, then a value of $y(\sqrt{2})$ is

(A) $\dfrac{\pi}{3\sqrt{2}}$ (B) $\dfrac{3}{2\sqrt{2}}$ (C) $\dfrac{1}{2\sqrt{2}}$ (D) $\dfrac{3\pi}{2\sqrt{2}}$

Q34. MCQ (Single Correct Answer)

Open solution

Let the area enclosed by the solution curve $y=y(x)$ of the differential equation $\dfrac{dy}{dx}+\dfrac{x-2}{y-b}=2$, $b>0$, $y(5)=0$ is 13π. Let $y=y(x)$ intersect the y-axis at the points P and Q. If the tangents to $y=y(x)$ at P and Q meet at T, then the area of ΔPTQ is

(A) 13.5 (B) 13 (C) 14.5 (D) 15

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Q35. MCQ (Single Correct Answer)

If foci of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{12} = 1$ and the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ coincide, then the eccentricity of the hyperbola, with foci $\left(0, \pm\sqrt{10}\right)$ passing through the point $(a, a+1)$, is

(A) 2 (B) $\sqrt{2}$ (C) $\frac{3}{\sqrt{5}}$ (D) $\sqrt{\frac{5}{2}}$

Open solution

Q36. MCQ (Single Correct Answer)

If the mean and median of the data

x	$0-10$	$10-20$	$20-30$	$30-40$	$40-50$	
f	3	6	2	x	y	$\sum f = 20$

are equal, then xy^2 is equal to

(A) 54 (B) 80 (C) 100 (D) 108

Open solution

Q37. Numerical

The number of solutions of the equation $|x^2 + 3x + 2| + |x + 5| - 2 = 0, x \in \mathbb{R}$, is _____.

Open solution

Q38. Numerical

If p_1 and p_2, respectively, are the second largest and the fourth largest prime numbers in the factorization of the coefficient of x^{30} in the expansion of $\left(x^2 + 2 + \frac{1}{x^2}\right)^{-5}(1 + x^2)^{40}$, then $p_1 + p_2$ is equal to _____

Open solution

Q39. Numerical

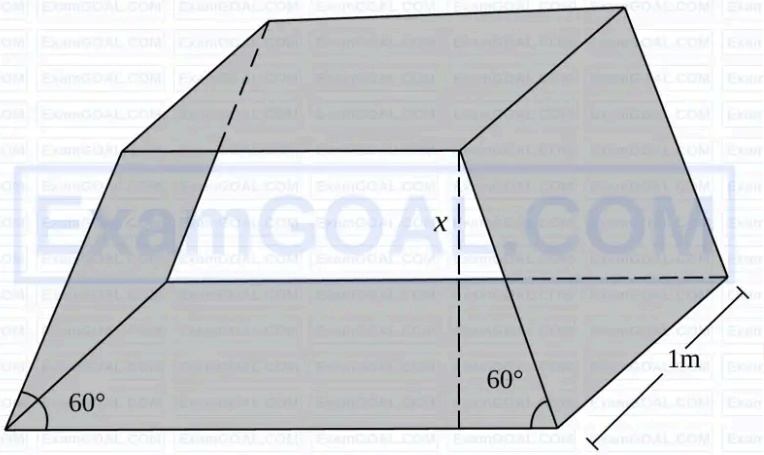
The number of integral values of b for which $2x^3 - 3x^2 - 12x + b = 0$ has three distinct real roots is _____.

Open solution

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Q40. Numerical

An air duct is made of a thick metallic sheet. The duct is open from front and back. Its cross-section is a trapezium of base angles 60° and area $1.5\ \mathrm{m}^2$. Its length is $1\ \mathrm{m}$. (see the figure). If its height $x = x_0$ (in metres) so that the area of the metallic sheet used for its construction is minimum, then $16(x_0)^4$ is equal to _____.



Open solution

Q41. Numerical

If the area enclosed by the curves $y = x^2$, $y^3 = x$, $x = -1$ and $y = x^2$, $y^3 = x$, $x = 1$, above the line $x = 2y$ is $\frac{n}{n+1}$, then n is equal to _____.

Open solution

Q42. Numerical

Let $A(0, a+2)$, $B(0, a)$, $C(-2, 0)$ and $D(2, 0)$ be four points and the lines AD and BC intersect at $P(x, y)$. If the locus of P is the curve $f(x, y) = 0$ and the tangent at the point $(4, \gamma)$ on this curve is $\frac{x}{\alpha} + \frac{y}{\beta} = 1$, then $\alpha(\beta - \gamma)$ is equal to _____.

Open solution

Q43. Numerical

Let L_1 be a line in the yz-plane with y and z intercepts $\frac{1}{4}$ and $\frac{1}{C}$ $(C > 0)$ respectively. Let L_2 be the line in the xz-plane with x and z intercepts $\frac{1}{3}$ and $-\frac{1}{C}$ respectively. If the shortest distance between L_1 and L_2 is $\frac{1}{5}$, then C^2 is equal to _____.

Open solution

Q44. Numerical

Let $OABC$ be a parallelogram, O be the origin, $A(2, 4, -5)$ and $C(b, 2, 3)$. If $P(a, a, a)$ and $Q(9 - a^2, 3, a - 1)$, $a \in \mathbb{N}$, are two points such that the projection of $\overrightarrow{OP}$ on $\overrightarrow{OB}$ is 2 and $\overrightarrow{OQ}$ makes acute angles with all three co-ordinate axes, then $\left|\overrightarrow{OB}\right|^2 + \left|\overrightarrow{AC}\right|^2$ is equal to _____.

Open solution

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Q45. Numerical

The probability distribution of a random variable X is $P(X=i)=\frac{1}{2^i}$, $i=1,2,3,\ldots$ The variance of X is equal to _____.

Open solution

Q46. Numerical

Let $S=\left\{\theta\in[0,2\pi]-\left\{\frac{\pi}{2},\frac{3\pi}{2}\right\}:\sin^{-1}(\sin\theta)+\cos^{-1}(\cos\theta)+\tan^{-1}(\tan\theta)=\frac{4\pi}{5}\right\}$.

Then $\frac{30}{\pi}\sum_{\theta\in S}\theta$ is equal to _____.

Open solution

Physics 28 questions

Q47. MCQ (Single Correct Answer)

If current (I), kinetic energy (K) and charge (Q) are taken as the fundamental quantities, The dimensional representation of power will be :

(A) $[\mathrm{K}^{-1}\,\mathrm{Q}\,\mathrm{I}^{-1}]$ (B) $[\mathrm{K}\,\mathrm{I}\,\mathrm{Q}^{-1}]$ (C) $[\mathrm{K}\,\mathrm{Q}\,\mathrm{I}^{-1}]$ (D) $[\mathrm{K}^{-1}\,\mathrm{Q}^{-1}\,\mathrm{I}]$

Open solution

Q48. MCQ (Single Correct Answer)

Two trucks A and B are approaching each other on a straight path with the velocities of $16\ \mathrm{m/s}$ and $20\ \mathrm{m/s}$ respectively. When they are $200\ \mathrm{m}$ apart their drivers see each other and applies the breaks simultaneously. If the truck A decelerates with $2\ \mathrm{m/s^2}$ and truck B decelerates with $4\ \mathrm{m/s^2}$, what is the distance between them when they finally stop?

(A) $14\ \mathrm{m}$ (B) $114\ \mathrm{m}$ (C) $86\ \mathrm{m}$ (D) $64\ \mathrm{m}$

Open solution

Q49. MCQ (Single Correct Answer)

A bullet of mass 10 gram is fired with velocity of $100\ \mathrm{m/s}$ from the gun of mass $1\ \mathrm{kg}$. Recoil speed of the gun is :

(A) $1\ \mathrm{m/s}$ (B) $10\ \mathrm{m/s}$ (C) $0.5\ \mathrm{m/s}$ (D) $0.1\ \mathrm{m/s}$

Open solution

Q50. MCQ (Single Correct Answer)

A mass of $0.5\ \mathrm{kg}$ is attached to one end of a massless spring whose natural length is $1\ \mathrm{m}$ and spring constant is $800\ \mathrm{N/m}$ The other end of the spring is fixed while the mass moves in a circular path in a horizontal plane with an angular speed of $20\ \mathrm{rad/s}$. The extension in the length of the spring will be :

(A) $\frac{1}{10}\ \mathrm{m}$ (B) $\frac{1}{4}\ \mathrm{m}$ (C) $\frac{1}{3}\ \mathrm{m}$ (D) $\frac{1}{5}\ \mathrm{m}$

Open solution

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Q51. MCQ (Single Correct Answer)

The time period of X planet of our solar system is 8 years. The distance of earth from the sun is $1.5 \times 10^{11}\ \mathrm{m}$. The distance of the X planet from the sun will be :

(A) $3\sqrt{2} \times 10^{11}\ \mathrm{m}$ (B) $3 \times 10^{11}\ \mathrm{m}$

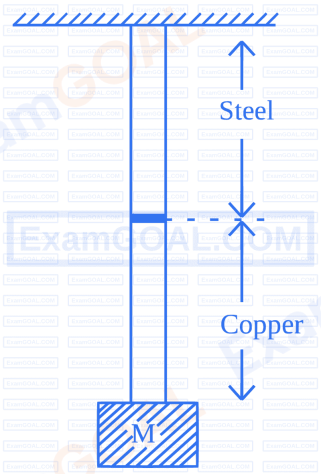
(C) $6 \times 10^{11}\ \mathrm{m}$ (D) $12 \times 10^{11}\ \mathrm{m}$

Open solution

Q52. MCQ (Single Correct Answer)

A block of mass M is attached to a wire. The upper part of wire is made of steel and lower part of wire of copper (as shown in figure). Both parts have the same area of cross section. Neglecting the mass of wires, the ratio of longitudinal strain developed in steel and copper wires is :

(Given, Young's modulus of steel $= 2 \times 10^{11}\ \mathrm{N/m^2}$ and Young's modulus of copper $= 1.5 \times 10^{11}\ \mathrm{N/m^2}$)



Open solution

(A) $\frac{3}{5}$ (B) $\frac{4}{5}$ (C) $\frac{4}{3}$ (D) $\frac{3}{4}$

Q53. MCQ (Single Correct Answer)

If the r.m.s speed of oxygen at room temperature is $\sqrt{56}\ \mathrm{m/s}$. The r.m.s speed of nitrogen at the same temperature is _____.

(A) $8\sqrt{2}\ \mathrm{m/s}$ (B) $8\ \mathrm{m/s}$ (C) $16\ \mathrm{m/s}$ (D) $7\ \mathrm{m/s}$

Open solution

Q54. MCQ (Single Correct Answer)

An oscillating simple pendulum of time period T_o is placed in a lift which is accelerating upwards with $2.5\ \mathrm{m/s^2}$. The time period of pendulum in the lift will be (if $g = 10\ \mathrm{m/s^2}$)

(A) T_o (B) $\frac{2}{\sqrt{5}} T_o$ (C) $\frac{2}{\sqrt{3}} T_o$ (D) $\frac{T_o}{4}$

Open solution

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Q55. MCQ (Single Correct Answer)

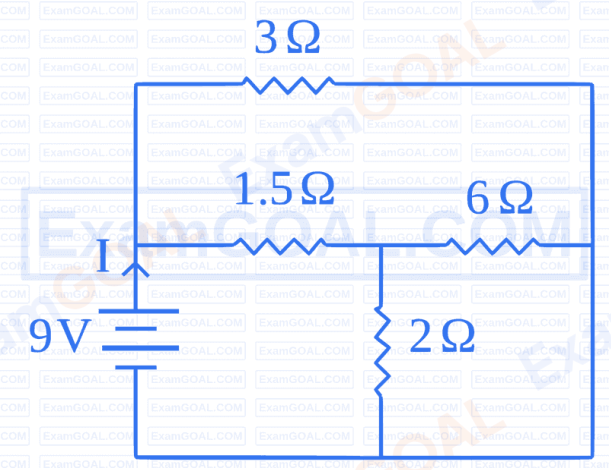
A charge q is uniformly distributed along the length L of a rod. It is then bent in the shape of a semicircle. The magnitude of electric field at the centre of semicircle will be :

(A) $\frac{q}{2\epsilon_0 L^2}$ (B) $\frac{q}{2\pi\epsilon_0 L^2}$ (C) $\frac{q}{2\pi\epsilon_0 L}$ (D) $\frac{q}{4\pi^2\epsilon_0 L}$

Open solution

Q56. MCQ (Single Correct Answer)

The current I drawn out of the battery connected in the given circuit is:



(A) 5 A (B) 6 A (C) 7 A (D) 8 A

Open solution

Q57. MCQ (Single Correct Answer)

In an A.C circuit, V & I are given by $V = 250\sin(100t)$ volt and $I = 10\sin\left(100t + \frac{\pi}{3}\right)$ A respectively

The power dissipated in the circuit is:

(A) 2500 W (B) 625 W (C) 1250 W (D) $625\sqrt{2}$ W

Open solution

Q58. MCQ (Single Correct Answer)

A plane electromagnetic wave propagating in free space has amplitude of electric field of $900\ \frac{\text{N}}{\text{C}}$.

The amplitude of magnetic field will be :_____

(A) $9\ \mu\text{T}$ (B) $27\ \mu\text{T}$ (C) $3\ \mu\text{T}$ (D) $270\ \text{GT}$

Open solution

Q59. MCQ (Single Correct Answer)

Two poloroids P_1 and P_2 are placed parallel to each other with common axis.The light of intensity I_0 passes through a polaroid sheet P_1 and then passes through polaroid P_2. If P_2 is now rotated by $60°$, then the intensity of the output light from P_2 will be:

(A) $\frac{I_0}{2}$ (B) $\frac{I_0}{4}$ (C) Zero (D) $\frac{I_0}{8}$

Open solution

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Q60. MCQ (Single Correct Answer)

A wire of length $8\ \mathrm{m}$ is bent to form a circular loop in y-z plane. A current of $0.5\ \mathrm{A}$ is flowing in anticlockwise direction in it, the magnetic dipole moment of the loop is:

(A) $-8\,\hat{i}\ \mathrm{A\,m^2}$

(B) $\frac{8}{\pi}\,\hat{i}\ \mathrm{A\,m^2}$

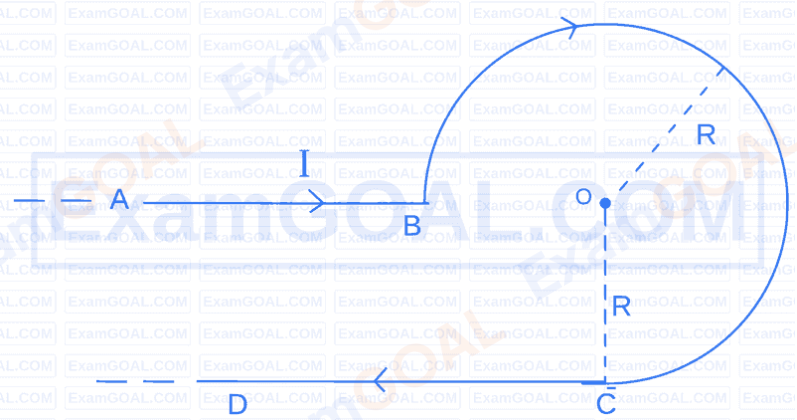
(C) $-4\pi\,\hat{i}\ \mathrm{A\,m^2}$

(D) $4\pi\,\hat{i}\ \mathrm{A\,m^2}$

Open solution

Q61. MCQ (Single Correct Answer)

In the following loop, magnitude of the magnetic field produced at the centre (O) of the loop is : (Assume AB and CD are infinitely extended):



(A) $\frac{\mu_0 I}{4\pi R}\left(\frac{3}{2}\pi\right)$

(B) $\frac{\mu_0 I}{2\pi R}$

(C) $\frac{\mu_0 I}{4\pi R}\left(\frac{3}{2}\pi+1\right)$

(D) $\frac{\mu_0 I}{4\pi R}\left(\frac{3}{2}\pi-1\right)$

Open solution

Q62. MCQ (Single Correct Answer)

Using Boh'r model of quantization of angular momentum the relation between the radius 'r' of the nth allowed orbit of quantum number 'n' for an electron in hydrogen atom is:

(A) $r \propto n^{\frac{1}{2}}$

(B) $r \propto n^2$

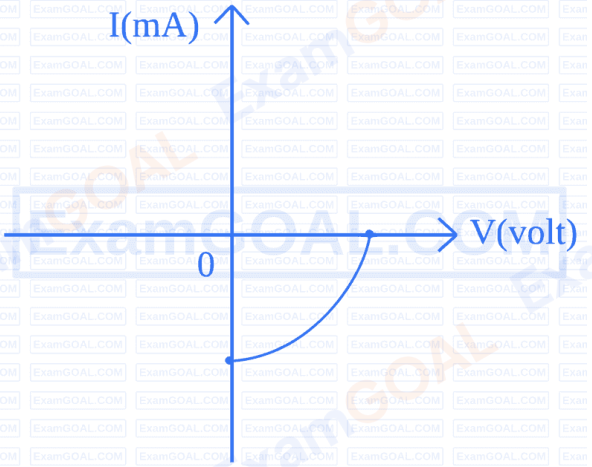
(C) $r \propto \left(\frac{1}{n}\right)^2$

(D) $r \propto \left(\frac{1}{n}\right)^{\frac{1}{2}}$

Open solution

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Q63. MCQ (Single Correct Answer)



Open solution

The I-V characteristics given in above figure is related to:

(A) Photodiode in reverse bias
(B) LED in forward bias
(C) Zener diode
(D) Solar cell

Q64. MCQ (Single Correct Answer)

The de Broglie wavelength associated with an electron, accelerated through a potential difference of $15056\ \text{V}$, is:

(A) 0.1 Å
(B) 1 Å
(C) 10 Å
(D) 100 Å

Open solution

Q65. Numerical

A nuclear physicist performed an experiment on the scattering of alpha particles by thin metal foil of gold. The alpha particles were accelerated up to kinetic energy 3.2 Mev. The estimation of the radius of the nucleus of gold element ($Z = 79$) by him is _____ $\times 10^{-16}\ \text{m}$

(Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9\ \frac{\text{N m}^2}{\text{C}^2}$)

Open solution

Q66. Numerical

A ring is placed at the bottom of a trough containing four immiscible liquids of refractive indices 1.0, 2.0, 3.0 and 4.0 poured one above the other of heights $10\ \text{cm}$, $20\ \text{cm}$, $30\ \text{cm}$, and $40\ \text{cm}$ respectively. The apparent depth at which the ring appears, when seen from outside is _____ cm.

Open solution

Q67. Numerical

A sinusoidal voltage $V(t) = 200 \sin 2000t$ volt is applied to a series LCR circuit in which $L = 10\ \text{mH}$, $C = 25\ \mu\text{F}$ and $R = 100\ \Omega$. The impedance of the circuit is _____ Ω

Open solution

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Q68. Numerical

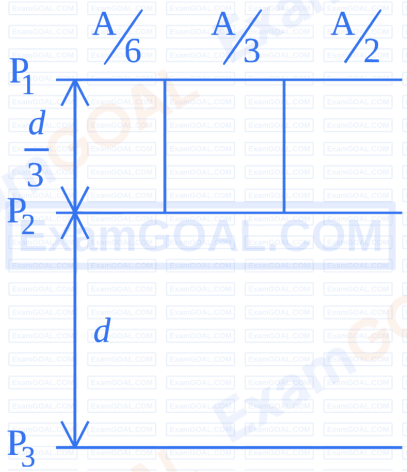
A cell sends a current through a resistance $4\ \Omega$ for time t and sends current through another resistance $16\ \Omega$ for the same time t. If the same amount of heat is developed in both the resistances, then the internal resistance of the cell is _____ Ω

Open solution

Q69. Numerical

Combination of four parallel plate air capacitors is shown in figure. Separation between the plate P_1 & P_2 is $\frac{d}{3}$ and separation between the plates P_2 and P_3 is d. The equivalent capacitance of the configuration is _____ μF.

(Given $\frac{\epsilon_0 A}{d} = 4\ \mu F$. Where A=Area of plates)



Open solution

Q70. Numerical

A particle executing SHM has its velocity $20\ cm\ s^{-1}$ at mean position and acceleration of $25\ cm\ s^{-2}$ at one of its extreme position. The amplitude of the particle will be ____ cm.

Open solution

Q71. Numerical

A water drop of radius $1\ cm$ is broken into eight equal droplets. Surface tension of water is $0.075\ N\ m^{-1}$. The gain in surface energy is _____ $\times 10^{-7}$ J. (Take $\pi = 3.14$)

Open solution

Q72. Numerical

A fly wheel of mass $10\ kg$ and radius $50\ cm$ is rotating at a rate of 360 rpm. Assuming mass to be concentrated at the rim, the constant retarding torque required to stop the rotation of wheel in 6 rotations will be $x\pi\ N\ m$. The value of x is _____

Open solution

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Q73. Numerical

A body moving horizontally on a smooth surface with a speed of $20\ \mathrm{m\,s^{-1}}$ splits into two parts and continue to move in the same direction . The masses of two parts are in the ratio of $1:2$. The smaller part moves with a speed of $30\ \mathrm{m\,s^{-1}}$. The fractional change is the kinetic energy is given by $\frac{1}{x}$. The value of x is_____

Open solution

Q74. Numerical

Two projectiles are thrown towards each other at 15° and 45° angles respectively with the horizontal at the same speed. The difference between the horizontal distances traveled by the two projectiles is $80\ \mathrm{m}$. The initial speed of the projectiles is _____ $\mathrm{m\,s^{-1}}$

[Given $g = 10\ \mathrm{m\,s^{-2}}$]

Open solution

Out of syllabus 16 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q75. MCQ (Single Correct Answer) · **Chemistry**

Combination of temperature and pressure causing the greatest deviation from ideal gas behaviour is:

(A) $100^\circ\ \mathrm{C}$ and $8\ \mathrm{atm}$

(B) $100^\circ\ \mathrm{C}$ and $4\ \mathrm{atm}.$

(C) $-100^\circ\ \mathrm{C}$ and $8\ \mathrm{atm}$

(D) $0^\circ\mathrm{C}$ and $4\ \mathrm{atm}$

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Freundlich's adsorption isotherm (in terms of concentration) is given by the expression; $\frac{x}{m} = K \cdot C^{1/n}$. The possible conclusions that can be drawn from the expression are :

(A) When $1/n = 1$, the adsorption is directly proportional to concentration.

(B) When $1/n = 0$, the adsorption is independent of concentration.

(C) When $n = 0$, a plot of x/m vs C is a line parallel to x-axis

(D) When $n = 0$, a plot of x/m vs C is a curve.

Choose the correct option from the following :

(A) (A) and (B) only

(B) (B) and (D) only

(C) (B), (C) and (D) only

(D) (A), (B) and (C) only

Open solution

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Q77. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match **List I** with **List II**

List I	List II
A. Cyanide Process	I. Vapour phase refining
B. Froth flotation process	II. Aluminium
C. Hall-Heroult process	III. Dressing of ZnS
D. Mond process	IV. Extraction of Au

Choose the correct answer from the options given below :

(A) A-IV, B-III, C-II, D-I

(B) A-I, B-II, C-III, D-IV

(C) A-II, B-III, C-IV, D-I

(D) A-III, B-II, C-IV, D-I

Q78. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The hardness of water is generally expressed in terms of $CaCO_3$ equivalent. The possible reasons are as follows:

(A) Its molar mass is 100, so the calculations become easier.

(B) It decomposes at a temperature, 1200 K.

(C) It is insoluble in water.

(D) It is moisture insensitive.

Choose correct option from the following for the reasons.

(A) (A) and (B) only

(B) (A) and (D) only

(C) (B) and (D) only

(D) (A), (B), and (C) only

Q79. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Which of the following is not a component of photochemical smog?

(A) Ozone

(B) Peroxyacetyl nitrate

(C) Nitric oxide

(D) Sulphur dioxide

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Q80. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements:

Statement I: In Nylon 6, the monomer unit is caprolactam.

Statement II: The monomer unit in Nylon 6 is synthesized from cyclohexanone.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Q81. MCQ (Single Correct Answer) · **Mathematics**

Open solution

Let A be a matrix and $\begin{bmatrix}1 & 2\\0 & 1\end{bmatrix}A\begin{bmatrix}2 & 0\\-1 & 1\end{bmatrix}=\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$. If $|A|$ and $|A^2|$ are the roots of the quadratic equation $ax^2+bx+3=0$, then $a+b-ab$ is equal to

(A) -34 (B) 50 (C) -390 (D) 438

Q82. MCQ (Single Correct Answer) · **Mathematics**

Open solution

The coefficient of x^{1011} in the expression $(1+2x)^{2022}+2x(1+2x)^{2021}+(2x)^2(1+2x)^{2020}+\ldots+(2x)^{2022}$ is

(A) ${}^{2022}C_{1011}$ (B) ${}^{2022}C_{1011}\times(2)^{1011}$

(C) ${}^{2023}C_{1011}\times(2)^{1011}$ (D) ${}^{2023}C_{1011}$

Q83. MCQ (Single Correct Answer) · **Mathematics**

Open solution

$2\sum_{n=2}^{\infty}\frac{n(2n^2+3)}{(n+1)!}$ is equal to

(A) $2(e+7)$ (B) $2e+1$ (C) $2e+9$ (D) $2(e+1)$

Q84. MCQ (Single Correct Answer) · **Mathematics**

Open solution

$\lim_{n\to\infty}\sum_{k=1}^{n}\frac{n(n^2+k^2)}{(n^4+n^2k^2+k^4)}$ is equal to

(A) $\frac{\pi}{\sqrt{3}}$ (B) $\frac{2\pi}{3\sqrt{3}}$ (C) $\frac{\pi}{2\sqrt{3}}$ (D) $\frac{\pi}{3}$

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Q85. MCQ (Single Correct Answer) · **Mathematics**

Let the curves $y^2 = kx$ and $xy = -1$ have a common tangent, whose slope is $\frac{1}{2}$. Then k can **NOT** lie in the interval

(A) $(2, 4]$ (B) $(-3, -1)$ (C) $[-4, -3)$ (D) $(1, 3)$

Open solution

Q86. MCQ (Single Correct Answer) · **Mathematics**

Let $ax + by + cz + 3 = 0$ be the plane which bisects the acute angle between the planes $2x - y - 2z + 3 = 0$ and $3x - 2y + 6z + 8 = 0$. Then $a + b + c$ is equal to

(A) -31 (B) 28 (C) $\frac{14}{15}$ (D) -28

Open solution

Q87. MCQ (Single Correct Answer) · **Mathematics**

The angle of elevation of the top P of a vertical tower for a person standing at a point A on the horizontal ground due north of the tower is 45°. Another person B is standing $50\ \text{m}$ west of A on the ground. If the angle of elevation of P at B is 30° then the height (in meters) of the tower is

(A) $25\sqrt{2}$ (B) $50\sqrt{2}$ (C) $25\sqrt{6}$ (D) $\frac{50}{\sqrt{3} - 1}$

Open solution

Q88. MCQ (Single Correct Answer) · **Mathematics**

Which of the following is a contradiction?

(A) $((\sim p) \vee q) \Rightarrow \sim (p \Rightarrow q)$ (B) $(\sim (p \Rightarrow q)) \wedge (\sim p)$

(C) $(p \Rightarrow q) \wedge p$ (D) $(\sim p) \wedge (\sim q)$

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

A Carnot engine is operating between the two reservoirs of temperature $227\ ^\circ\text{C}$ and $127\ ^\circ\text{C}$ respectively. If the engine performs $1.04 \times 10^5\ \text{J}$ work per cycle. Then, the heat absorbed from hot reservoir per cycle will be :

(A) $2.08 \times 10^4\ \text{J}$ (B) $1.04 \times 10^5\ \text{J}$ (C) $5.20 \times 10^5\ \text{J}$ (D) $2.36 \times 10^5\ \text{J}$

Open solution

Q90. MCQ (Single Correct Answer) · **Physics**

An amplitude modulate wave consists of following components :

Carrier component $= 5\ \text{V}$ peak value

Lower side band component $= 2.5\ \text{V}$ peak value

Upper side band component $= 2.5\ \text{V}$ peak value

The amplitude of modulating signal is:

(A) $1\ \text{V}$ (B) $1.25\ \text{V}$ (C) $2.5\ \text{V}$ (D) $5\ \text{V}$

Open solution

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Answer Key

Chemistry

1. **A**	2. **A**	3. **B**	4. **B**	5. **C**
6. **B**	7. **B**	8. **A**	9. **B**	10. **A**
11. **D**	12. **B**	13. **C**	14. **B**	15. **4**
16. **3**	17. **271**	18. **42**	19. **1**	20. **3**
21. **1**	22. **4**	23. **9**	24. **2**	

Mathematics

25. **D**	26. **D**	27. **C**	28. **B**	29. **C**
30. **D**	31. *Scan QR*	32. **B**	33. **D**	34. **A**
35. **B**	36. **A**	37. **0**	38. **34**	39. **26**
40. **27**	41. **47**	42. **4**	43. **75**	44. **118**
45. **2**	46. **68**			

Physics

47. **B**	48. **C**	49. **A**	50. **C**	51. **C**
52. **D**	53. **B**	54. **B**	55. **A**	56. **B**
57. **B**	58. **C**	59. **D**	60. **B**	61. **C**
62. **B**	63. **D**	64. **A**	65. **711**	66. **40**
67. **100**	68. **8**	69. **3**	70. **16**	71. **942**
72. **15**	73. **8**	74. **40**		

Out of syllabus

75. **C**	76. **A**	77. **A**	78. **B**	79. **D**
80. **A**	81. **D**	82. **C**	83. *Scan QR*	84. **C**
85. **C**	86. **B**	87. **A**	88. **B**	89. **C**
90. **D**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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